



NSW Education Standards Authority

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Centre Number

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Student Number

2025 HIGHER SCHOOL CERTIFICATE EXAMINATION

Earth and Environmental Science

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- A Geological Time Scale is provided at the back of this paper
- Write your Centre Number and Student Number at the top of this page

Total marks:
100

Section I – 20 marks (pages 2–11)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 80 marks (pages 13–35)

- Attempt Questions 21–35
- Allow about 2 hours and 25 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

1 What did Urey and Miller produce in their experiments while investigating the origin of organic molecules?

- A. Amino acids
- B. Stromatolites
- C. Primitive cells
- D. Carbon dioxide

2 Australia has been drifting north, away from Antarctica, for about 85 million years.

What causes Australia and Antarctica to drift apart?

- A. Global warming
- B. Changes in ocean currents
- C. A divergent tectonic boundary
- D. Separation of the hydrosphere and cryosphere

- 3 The Volcanic Explosivity Index (VEI) is a scale for expressing the size of an explosive volcanic eruption. The data provides characteristics of eruptions for each VEI.

Volume of volcanic material emitted (m ³)	<10 ⁴	10 ⁴	10 ⁶	10 ⁷	10 ⁸	10 ⁹	10 ¹⁰	10 ¹¹	10 ¹²
How often?	Daily	Daily	Weekly	Yearly	10s of years	100s of years	100s of years	1000s of years	10 000s of years
VEI	0	1	2	3	4	5	6	7	8

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Based on the data above, what period of time would likely pass between two consecutive eruptions in the VEI 7 range?

- A. One day
 - B. One year
 - C. 100s of years
 - D. 1000s of years
- 4 Why is solid waste compacted in a landfill site?
- A. To speed up decomposition
 - B. To increase the space available
 - C. To reduce ongoing operational costs
 - D. To decrease the transport distance between facilities
- 5 Which of the following contributes most to the anthropogenic greenhouse effect?
- A. Fire stick farming
 - B. Changes in solar activity
 - C. Land clearing for agriculture
 - D. Increased use of rooftop solar panels

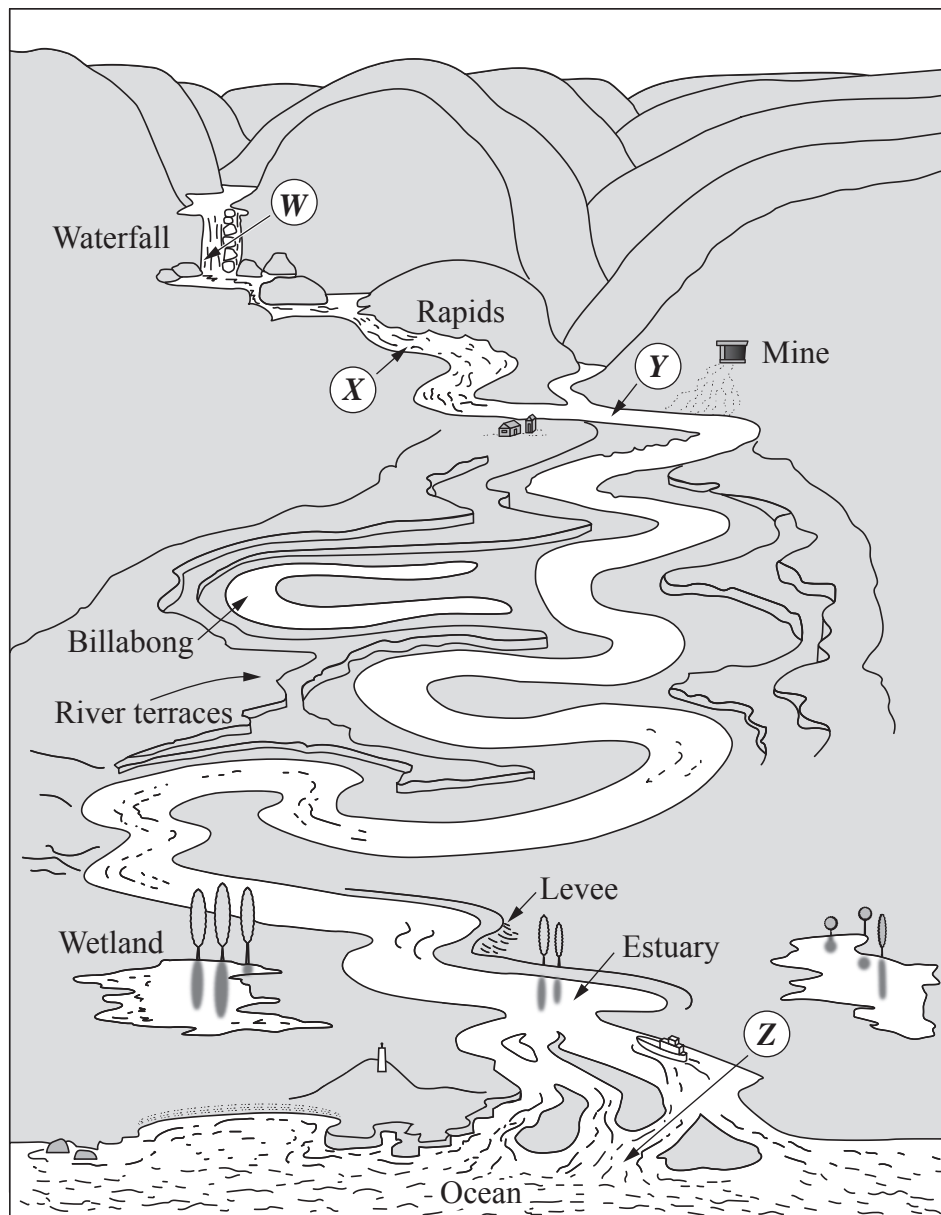
- 6 The image shows an imprint of a leaf in concrete. The process responsible for the leaf imprint is similar to the process responsible for the formation of mould fossils.



Which principle accounts for the similarity of the process involved in each case?

- A. Dendrochronology
- B. Extinction
- C. Superposition
- D. Uniformitarianism

- 7 The diagram shows a typical river system (not to scale).



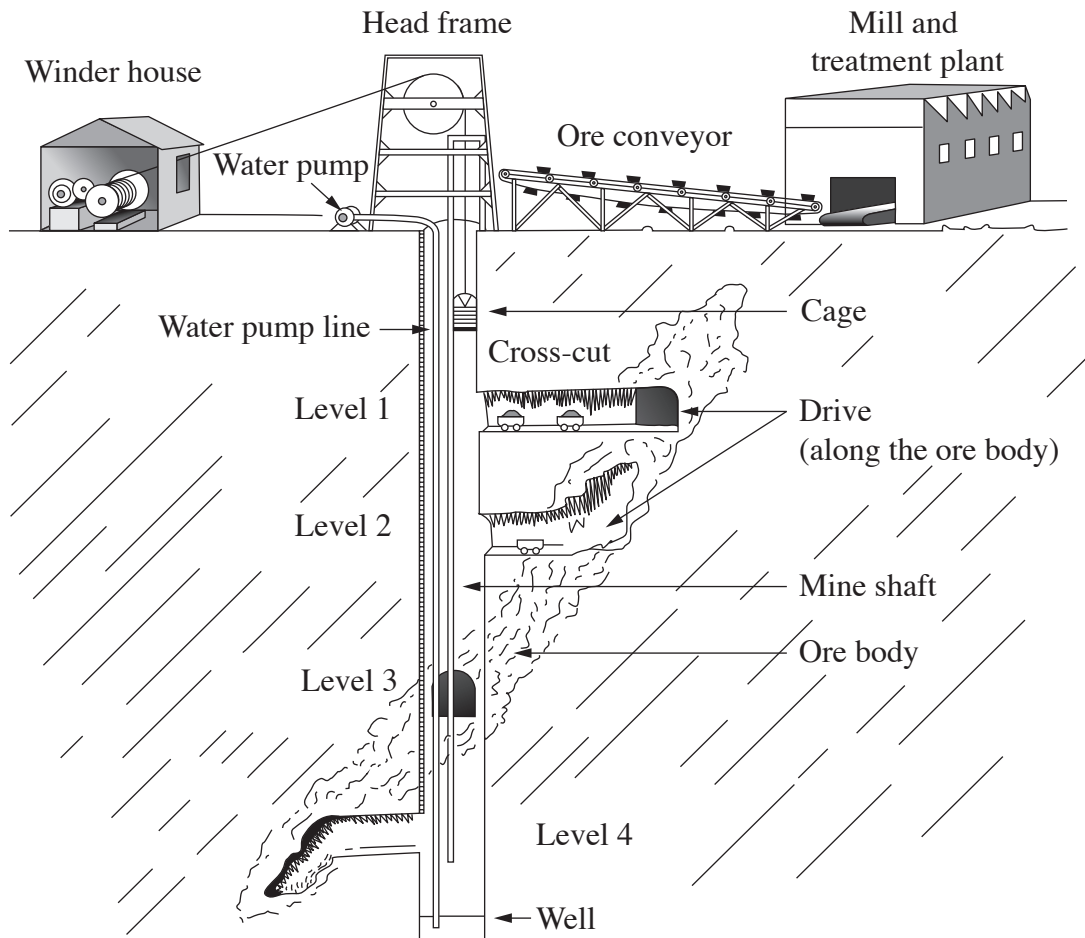
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Which location is most suitable for large-scale hydroelectric power production?

- A. W
- B. X
- C. Y
- D. Z

- 8** Which of the following is likely to contribute to high-speed lava flows?
- A. Relatively cool magma
 - B. Lava with high water content
 - C. Lava with relatively low silica content
 - D. High pressure within the magma chamber
- 9** What is the most likely effect of carbon dioxide being released from a single volcanic eruption?
- A. Local cooling
 - B. Rapid global warming
 - C. A minor increase in global warming
 - D. A drop in global temperatures over several months
- 10** A fossil found in NSW was formed during the Permian period.
- What is the most likely age of the fossil?
- A. 260 BP
 - B. 260 Ma
 - C. 320 BP
 - D. 320 Ma
- 11** A student wanted to collect data on the effectiveness of a sustainability initiative in their local community.
- Which of the following is the most accurate, reliable and valid source of information for this research?
- A. Reports about the initiative published by the project manager
 - B. A local television news report showing a protest against the initiative
 - C. An artificial intelligence (AI) generated analysis of Australian sustainability initiatives
 - D. Peer-reviewed scientific journal articles analysing international sustainability initiatives

- 12 The following image shows a mine which is reaching the end of its useful life. Once the mine is closed, the site should be reclaimed.



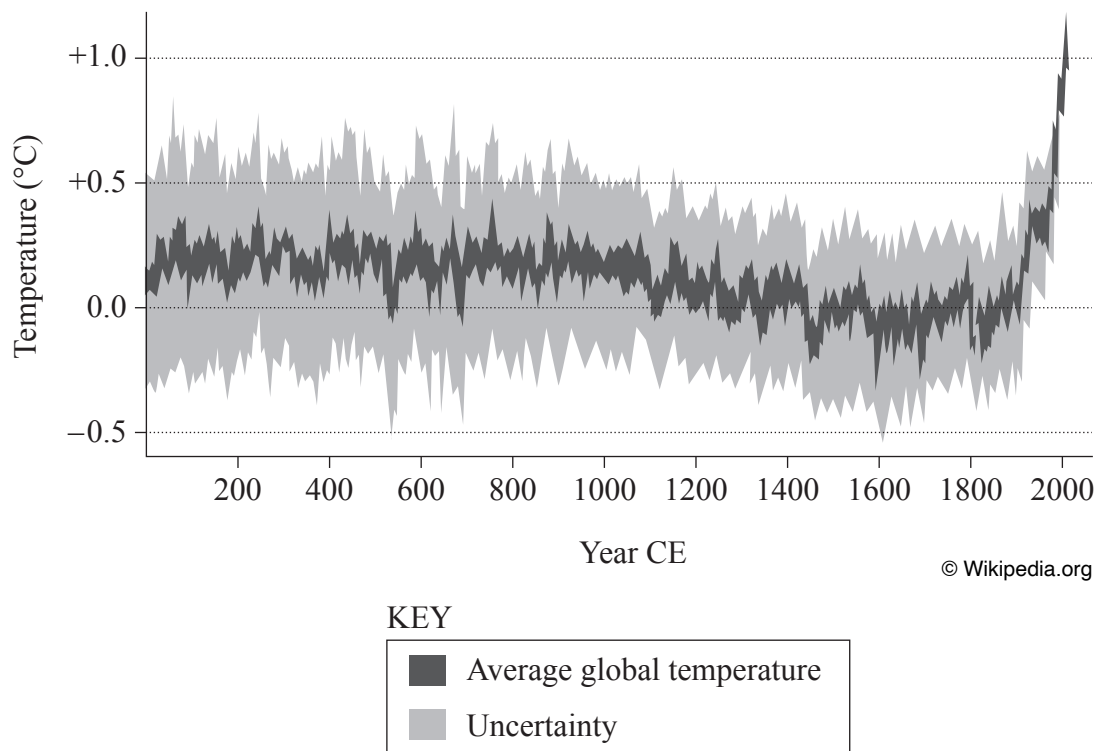
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What is the most appropriate reclamation strategy that could be used once mining has ceased and the structures have been removed?

- A. Backfill the mine shaft and fence the area
- B. Leave the shaft and drives open to provide animal access
- C. Build sporting fields around the mine shaft and use the water pump for irrigation
- D. Use explosives to collapse levels 1–4 to create a lake and wetland environment on the surface

- 13** Why is it difficult to make accurate long-term weather predictions?
- A. Meteorological models cannot keep up with rapid climate change.
 - B. The atmosphere is a dynamic system influenced by numerous variables.
 - C. Rainfall patterns are influenced by geographic features, which vary greatly across regions.
 - D. Long-range predictions are only based on historical data, which does not account for current conditions.
- 14** A student wants to conduct an experiment to test the effect of soil moisture on landslide frequency.
- Which of the following variables should be controlled?
- A. The type of soil used
 - B. The amount of soil displaced
 - C. The amount of water added to the soil
 - D. The time it takes for the soil to be displaced
- 15** Which of the following is derived from direct instrumental measurements of climate variation?
- A. The size of CO₂ bubbles trapped in ice reveal cyclical patterns of change.
 - B. All local temperatures in the future will be higher than current temperatures.
 - C. Average global temperatures have undergone cycles of change for the past 800 000 years.
 - D. There is a link between the concentration of greenhouse gases in the atmosphere and average global temperatures.

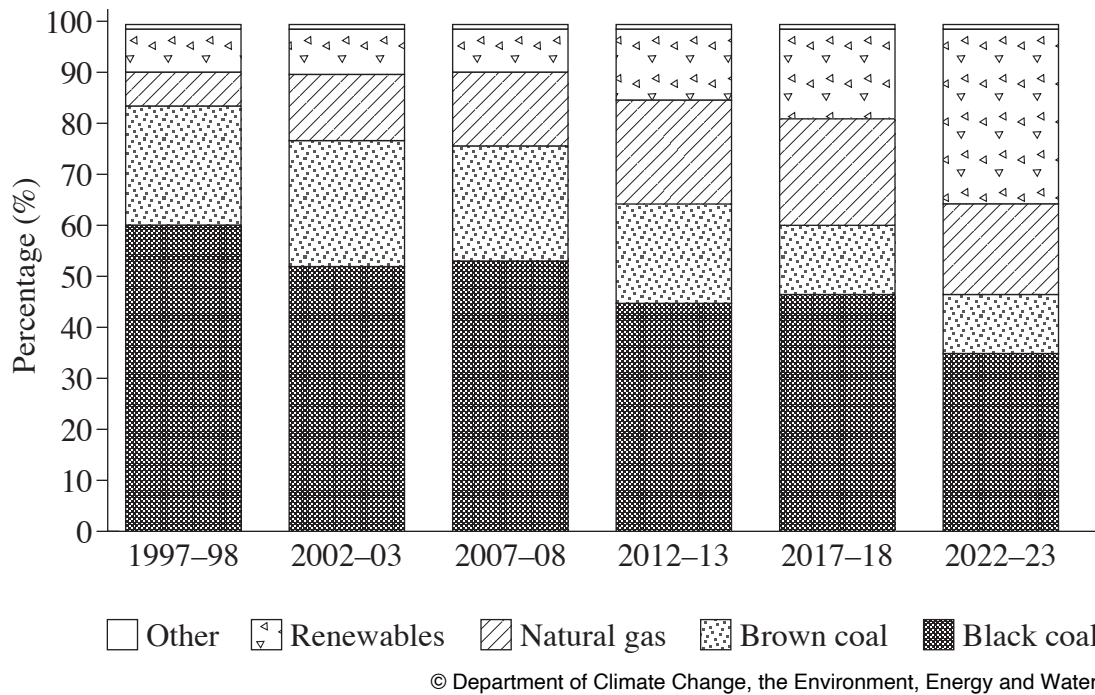
- 16 The graph shows the change in average global temperature over time.



Why does the estimated uncertainty in temperature on the graph become thinner over time?

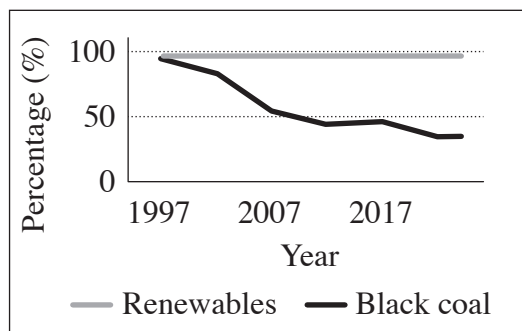
- A. Climate data have become more accurate.
 - B. There is a decrease in available evidence for dating.
 - C. The difference between polar and equatorial temperatures has decreased.
 - D. The difference between annual maximum and minimum temperatures has decreased.
- 17 Which statement best describes a key feature of the natural greenhouse effect?
- A. All solar energy, not absorbed by the atmosphere, is reflected into space.
 - B. Greenhouse gases in the atmosphere absorb and scatter infrared radiation.
 - C. Infrared radiation reflected from the Earth's surface cannot escape the atmosphere.
 - D. Depleted ozone levels allow increased ultraviolet radiation to be absorbed by greenhouse gases in the atmosphere.

- 18** The following graph shows the different fuel types used to produce electricity in Australia between 1997 and 2023.

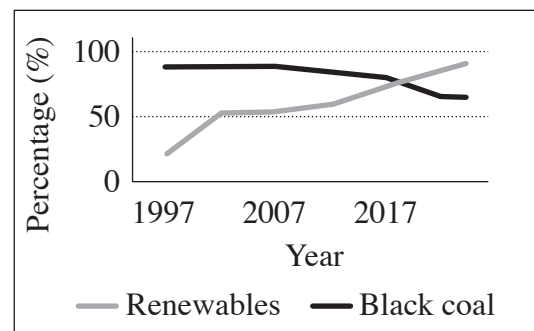


Which graph represents the trends for Australia's use of black coal and renewables between 1997 and 2023?

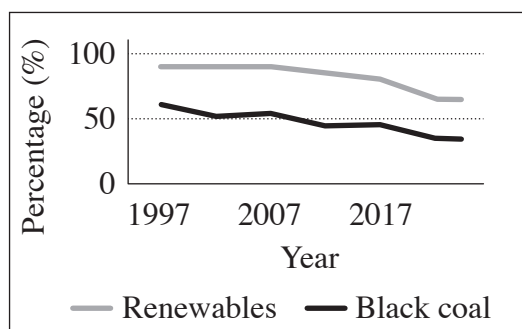
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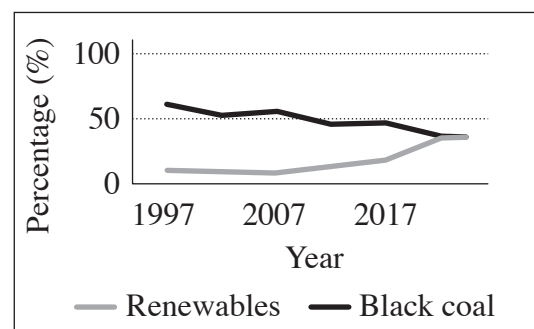
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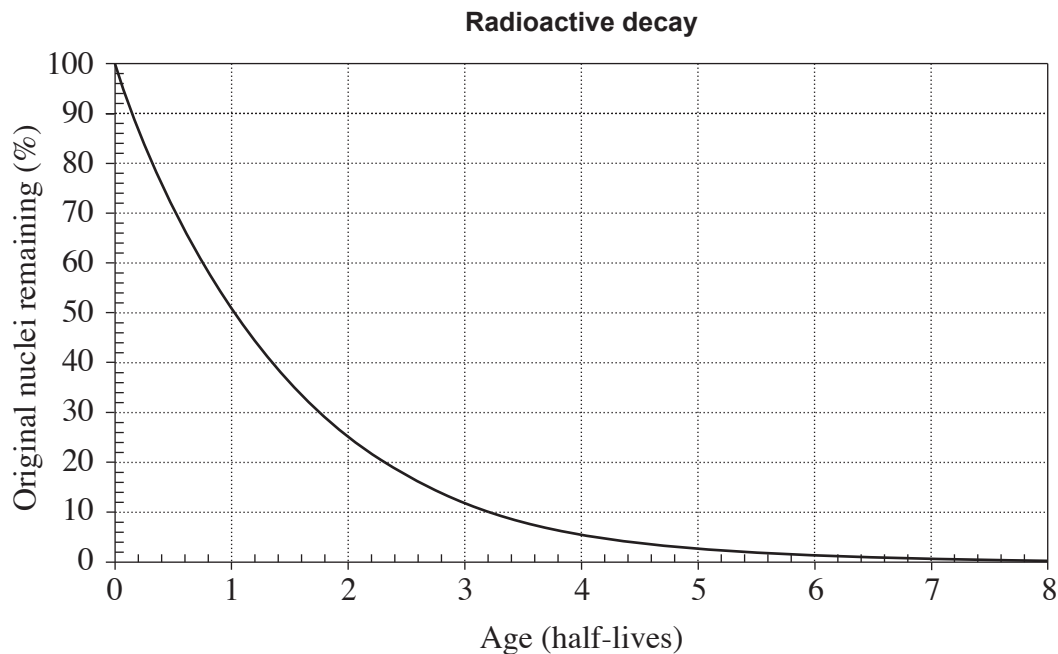
C.



D.



- 19 Which underlying principle allows scientists to use fossilised pollen grains to determine ancient climate?
- A. Fossilised pollen grains contain preserved DNA.
 - B. Plants have evolved over geological time in response to changing environmental conditions.
 - C. The climatic conditions that support different kinds of plants remain constant throughout time.
 - D. Microfossils such as pollen grains are the best source of data for determining past climatic conditions.
- 20 The age of igneous rocks can be determined using absolute dating methods, such as radiometric dating. Radioactive nuclei decay over time. The percentage of original radioactive nuclei remaining can be used to determine the age of a rock.



Uranium-235 decays to form lead-207 with a half-life of 704 million years. A sample was found containing 80% uranium-235 and 20% lead-207.

When did the sample form?

- A. 0.3 Ma BP
- B. 80 Ma BP
- C. Triassic period
- D. Proterozoic eon

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Centre Number

Earth and Environmental Science

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Student Number

Section II Answer Booklet

80 marks

Attempt Questions 21–35

Allow about 2 hours and 25 minutes for this section

Instructions

- Write your Centre Number and Student Number at the top of this page
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response
- Show all relevant working in questions involving calculations
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering

Please turn over

Question 21 (3 marks)

- (a) Name ONE resource that can be extracted from a mine.

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- (b) Outline ONE way that traditional owners can be involved in a large-scale mining operation.

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Question 22 (3 marks)

Describe ONE method of determining the age of a rock layer, other than absolute (radiometric) dating.

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Question 23 (3 marks)

Describe ONE resource management process used by Aboriginal and/or Torres Strait Islander Peoples.

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Question 24 (4 marks)

Compare EITHER the Deccan OR Siberian Traps eruptions with an eruption of Mt Pinatubo in driving natural climate change.

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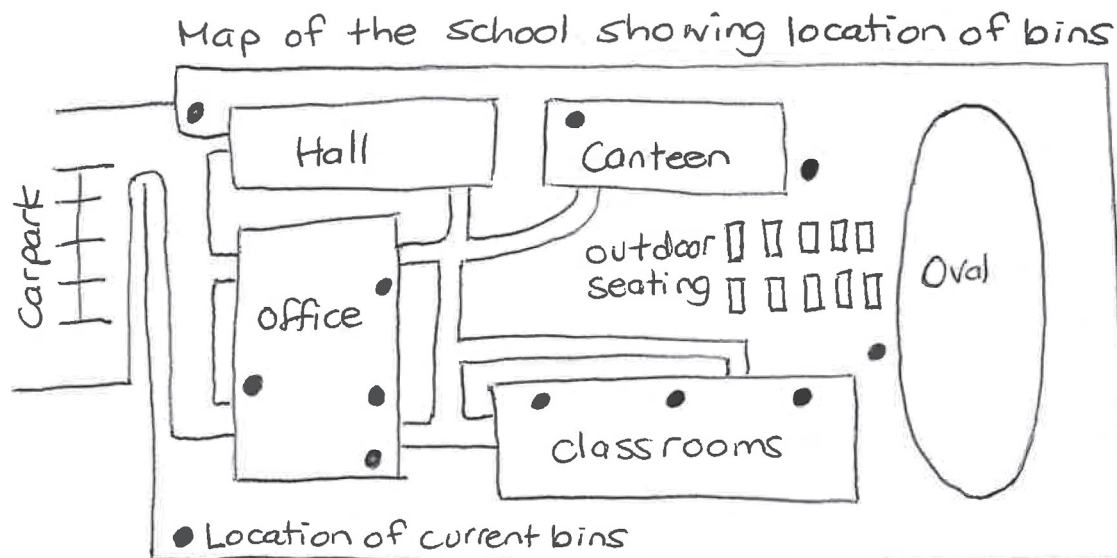
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Question 25 (8 marks)

A student leadership team has decided to take action to manage its school's waste. Currently, the entire contents of the school's bins are sent to landfill. A student drew a map of the school showing the location of the bins.



The team will perform a first-hand investigation and make recommendations to the school principal.

The aim of the investigation is to determine which areas of the school produce the greatest amount of recyclable waste.

- (a) Justify ONE safety measure the students should use when undertaking this practical investigation on waste.

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Question 25 continues on page 17

Question 25 (continued)

- (b) Write a valid method for an investigation to address the students' aim.

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- (c) As a result of the investigation, the school has approved recycling bins to reduce the school's contribution to landfill.

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Justify ONE location in the school where the placement of a recycling bin would be effective. Refer to the map in your answer.

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End of Question 25

Question 26 (7 marks)

Figure 1 and Figure 2 model the relative size of tsunami waves as they move closer to shore.

Figure 1

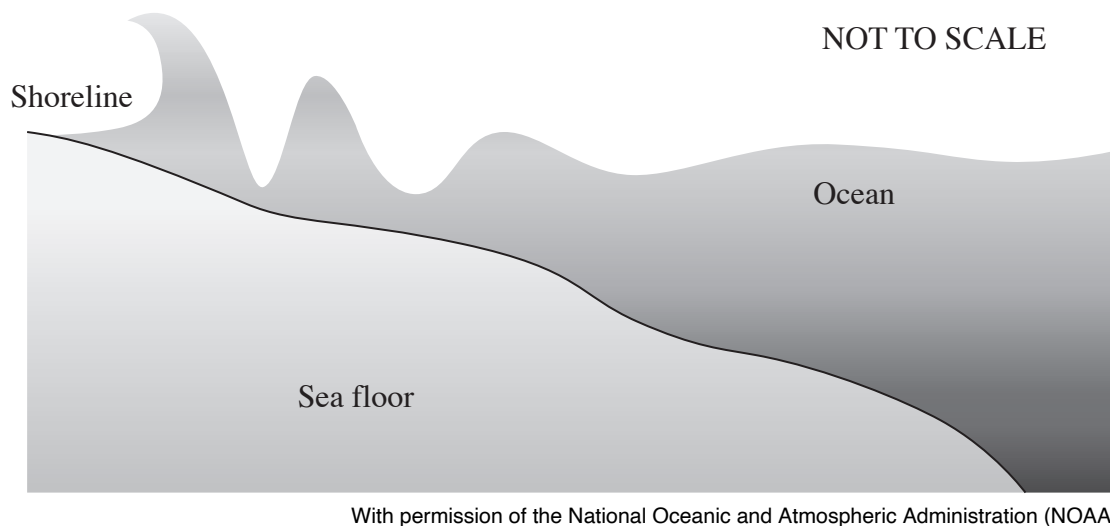
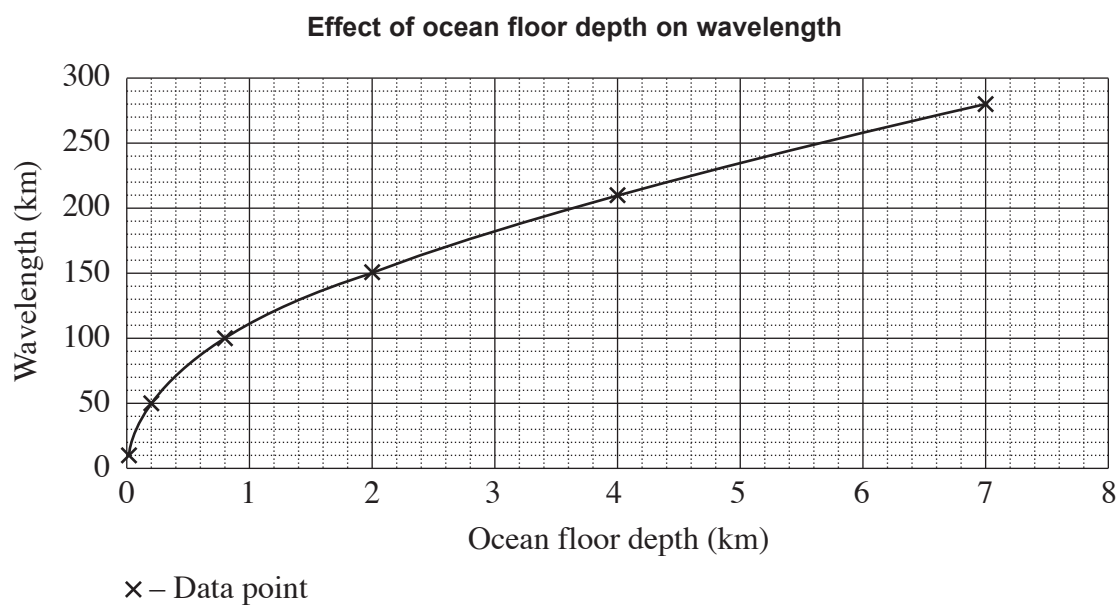


Figure 2

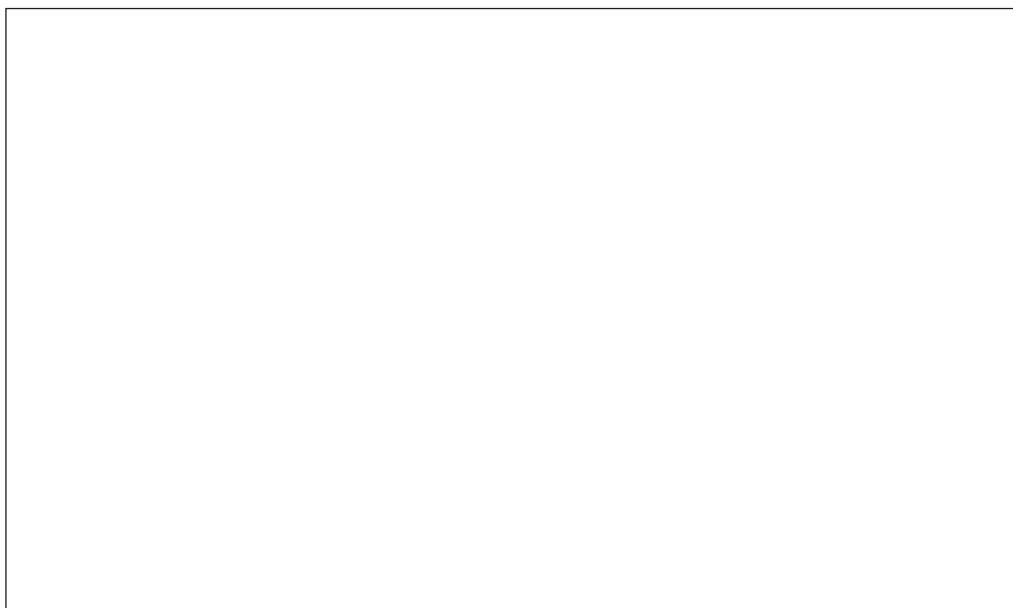


Question 26 continues on page 19

Question 26 (continued)

- (a) Construct a table to display the data shown in Figure 2. Include all data points (x) in your answer.

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- (b) Explain why tsunamis may become disasters when they reach the shoreline. Make reference to both figures to support your answer.

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End of Question 26

Question 27 (8 marks)

Evaluate the effectiveness of planting native trees in solving problems studied in the Earth and Environmental Science course. Support your answer with TWO specific, unrelated examples.

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Examples may include:

- natural disasters
- resource management problems
- climate related problems
- other environmental problems.

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Question 27 continues on page 21

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Question 28 (5 marks)

Assess the effect of TWO human activities on the sustainability of water resources.

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Question 29 (3 marks)

Contrast TWO effects of human activity and cyanobacteria on the atmosphere.

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Question 30 (3 marks)

The table shows the estimated timing of supercontinents throughout Earth's geological history.

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<i>Supercontinent</i>	<i>Formation (Ma BP)</i>	<i>Breakup (Ma BP)</i>
Columbia	1500	1400
Rodinia	900	750
Pangaea	300	180

Predict the timing for the formation of a future supercontinent. Use calculations to support your prediction.

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Question 31 (7 marks)

Individuals and scientific communities can help mitigate rising global temperatures.

- (a) Justify TWO actions that can be taken by an individual to minimise their impact on rising global temperatures. 3

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- (b) Explain ONE advantage and ONE disadvantage of a geo-engineering strategy that could be used to mitigate rising global temperatures. 4

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Question 32 (3 marks)

Explain how ONE technology used to monitor meteorological activity can be used to predict short-term extreme weather events.

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Please turn over

Question 33 (7 marks)

The ground shaking experienced by buildings during an earthquake is dependent on several factors including:

- displacement and type of fault movement
- depth of the focus (hypocentre)
- type of ground the building is constructed upon
- building design and construction.

(a) Explain why the focus of some earthquakes is deeper than others.

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Question 33 continues on page 27

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Question 33 (continued)

- (b) Analyse TWO factors which could affect the amount of damage a building experiences due to an earthquake. Include diagrams in your answer.

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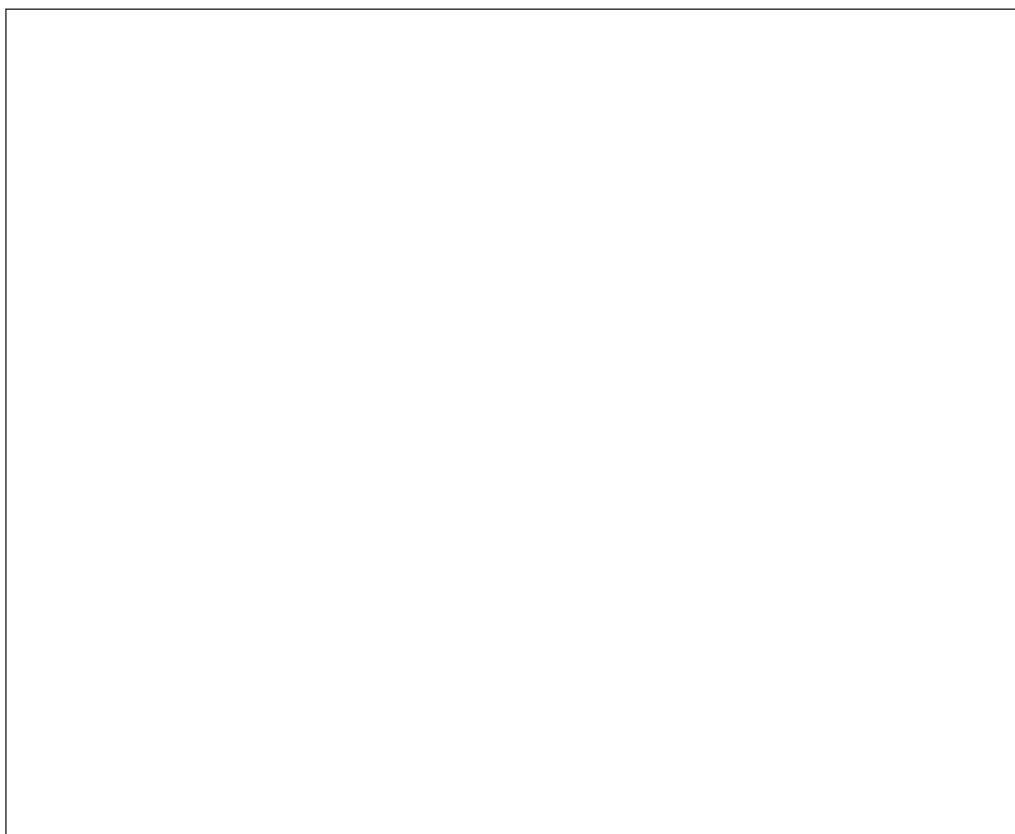
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End of Question 33

Question 34 (8 marks)

The supercontinent Rodinia broke apart due to processes involved in the plate tectonic supercycle, leading to the formation of the supercontinent Pangaea.

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Account for the breakup of Rodinia and the formation of Pangaea. Use cross-section diagrams to support your response.

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Question 34 continues on page 29

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Question 34 (continued)

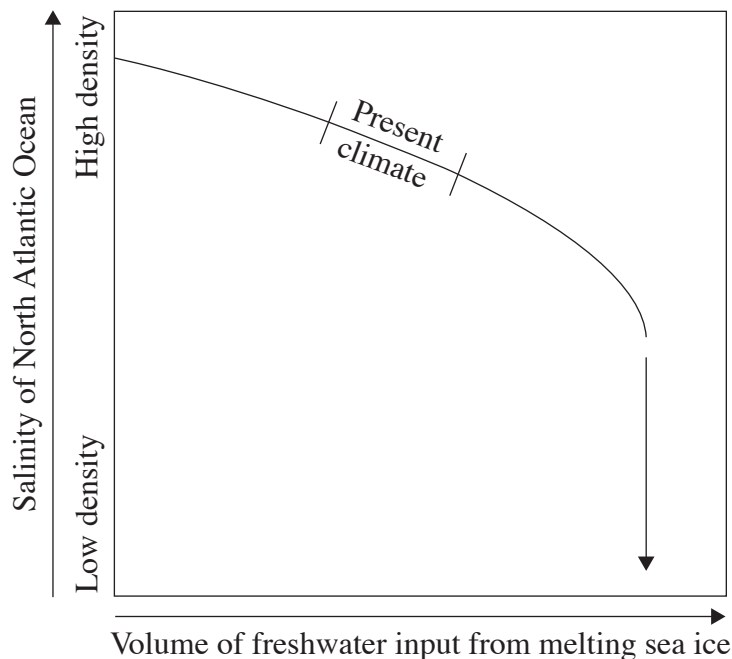
End of Question 34

Question 35 (8 marks)

The North Atlantic Deep Water (NADW) overturning current forms in the North Atlantic Ocean as warm salty water cools and sinks. It is one of the main drivers of the Great Ocean Conveyor Belt and influences the climate of Europe.

The graphs show the relationships between salinity, freshwater input and the strength of the NADW overturning current.

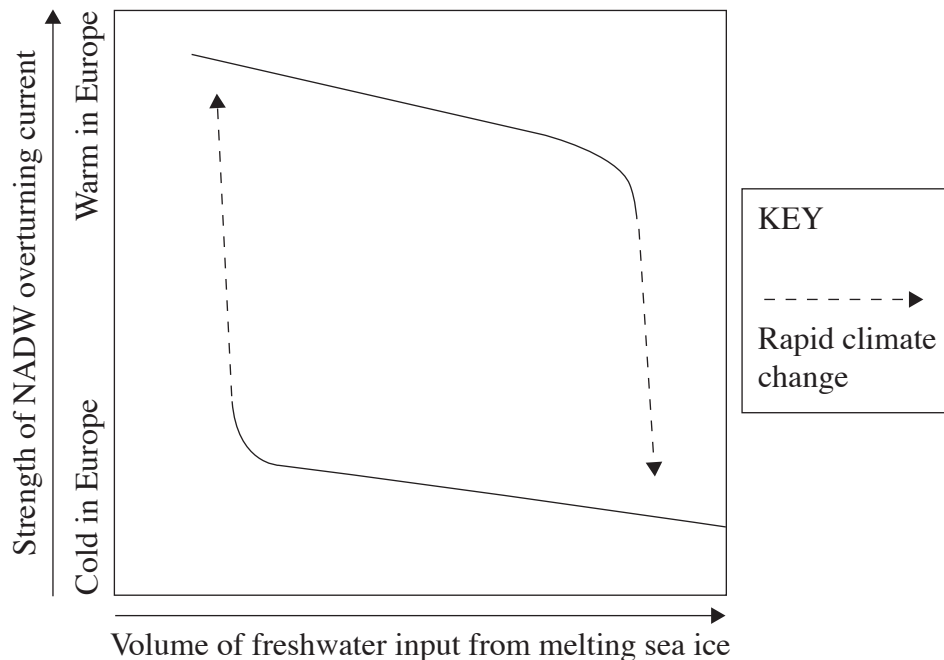
Graph 1



© PIK. S.R.

Courtesy of Professor Stefan Rahmstorf

Graph 2



KEY

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Rapid climate
change

© PIK. S.R.

Courtesy of Professor Stefan Rahmstorf

Question 35 continues on page 31

Question 35 (continued)

- (a) Use Graph 1 to outline the conditions in the North Atlantic Ocean under the present climate. 2

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- (b) Draw a flow chart to show how melting sea ice affects the NADW overturning current and the climate in Europe. Refer to both graphs in your answer. 4

- (c) Variations to the NADW overturning current can change climate in Europe. 2

Explain a likely flow-on effect of this change.

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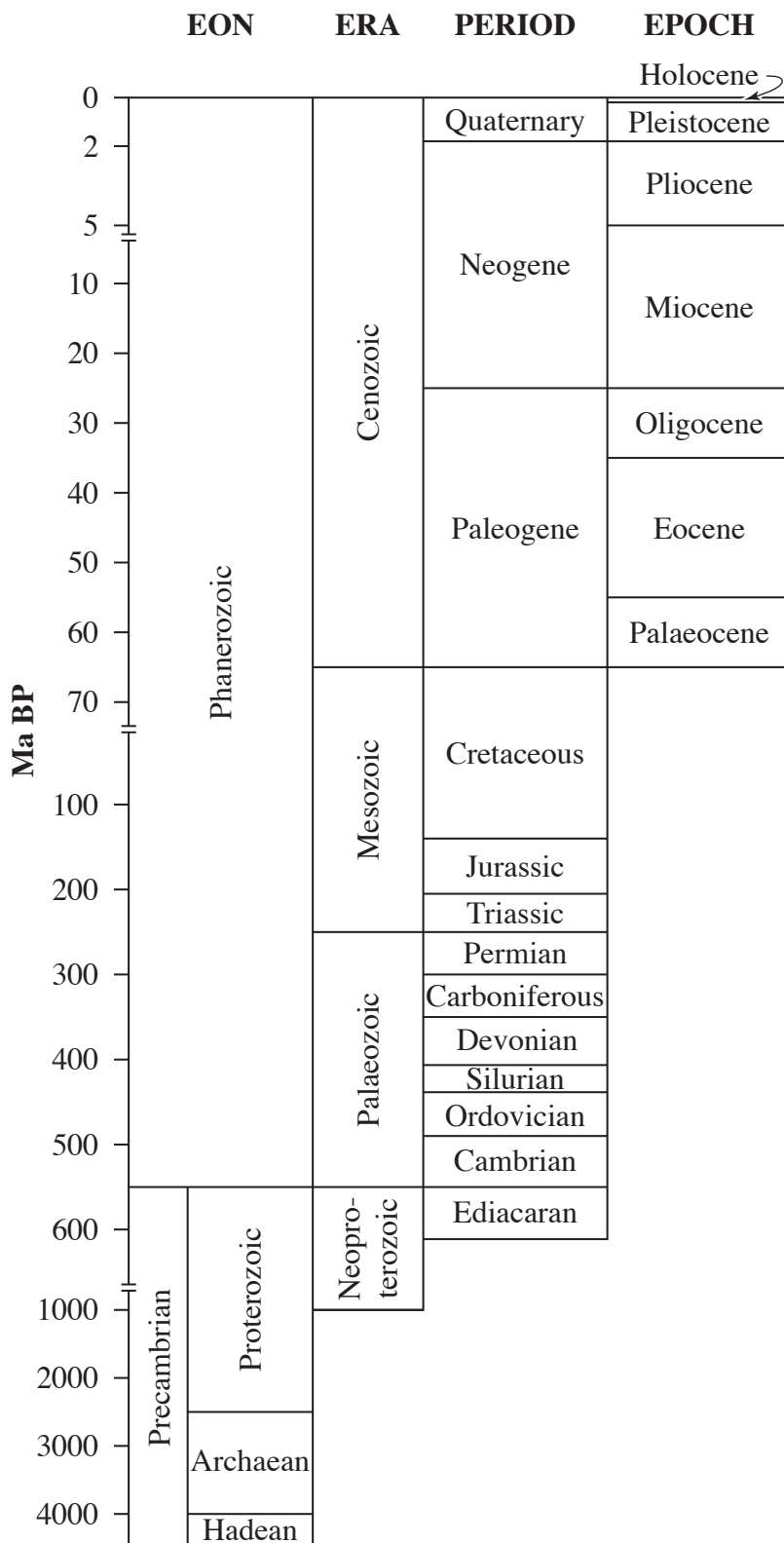
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Earth and Environmental Science

Geological Time Scale



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